

$$\bar{x} = 60$$

$$n = 36$$

$$\sigma = 6$$

$$\pi = 30$$

$$n = 36$$

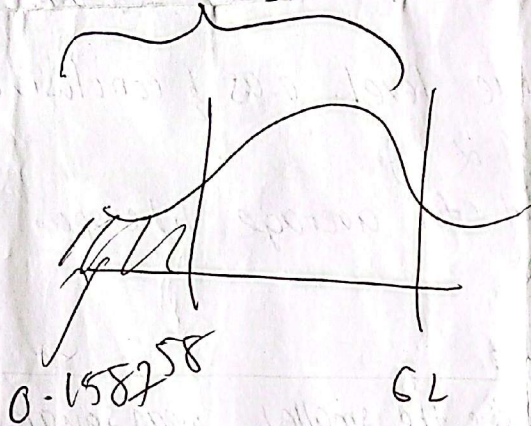
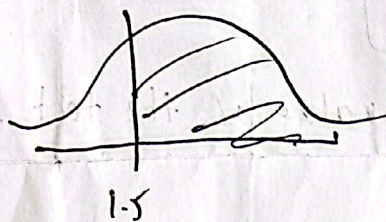
$$G = 6$$

$$\bar{c} = 11/2$$

$$2 > 1.5$$

$$58 < 2 < 62$$

$$0.8413$$



$$\frac{58 - 60}{2} = -1$$

$$0.8413$$

$$n = 1000$$

$$= 0.1587$$

mea

1) A company knows that average time to resolve a support ticket is 30 minutes, with S.D. 6. A manager selects 36 tickets, what's the probability that the sample mean resolution time is greater than 31.5

$$(0.9332)$$

$$50 = n$$

$$\bar{x} = 880$$

$$G = 50$$

2) A company processes shipment in avg of 60 minutes, with a s.d of 12 min. A random sample of 36 is selected. What's the probability that the sample mean is between 58 & 62.

$$(0.6826)$$

An estimator is said to be consistent if,

a) It always equal the parameter value.

b) Its variance is always zero.

c) It converges to the parameter value as the sample size increases.

d) It has a small margin of error.

e) None of the above.

$$(c)$$

Let α denote the level of significance in a two tail hypothesis test. The proportion of area covered by the critical region is equal to

a) α b) 2α c) $\alpha/2$ d) 0.05

e) None

1) A company manufactures light bulbs and claim that their light bulbs have an average lifespan of 1000 hrs. You, as a quality control manager, are skeptical about this claim & decide to conduct a ^{hypothesis} test. You collected a random sample of 50 light bulbs & sample mean is 980 hrs. and sample standard deviation is 50 hrs.

i) $H_0: \mu = 1000$ } state the null & alternative hypothesis
 $H_a: \mu \neq 1000$ }

ii) What is the value of the test statistic. (my value from $z = -0.0023$)

iii) Test the hypothesis with significance level 0.05 & conclusion.
ans (H_0 is rejected). ($0.0046 < \alpha$)

iv) Compute 95% confidence interval for average life span
(966.1407, 993.8593) //

2) a) An unbiased estimator will always have the smallest mean square error. (F)

b) A survey is conducted to estimate the average weight of students in a uni. The specific avg weight calculated from the sample is called the estimator (F)

c) The sampling distribⁿ of the sample mean always has the same shape as the populaⁿ distribⁿ (T)

d) A 95% confidence interval means there is a 95% probability the populaⁿ parameter is in the interval (F)

e) Increasing confidence interval makes the confidence interval wider. (T)

f) A Type I error occurs when we reject a true null hypothesis & true (T)

g) An estimator can be consistent even it is biased (T)

h) A p-value represents the probability that H_0 is true (T)

i) A 1-tail test uses all of the significance level (α) in one tail of the distribⁿ (T)

j) The t test is appropriate when populaⁿ variance is known (F)

1) $V = \{A, B, C, D, E, F, G, H\}$
 $E = \{(AB, 4), (AC, 7), (AD, 3), (AE, 8), (BC, 2), (BD, 6), (BF, 5), (CG, 4), (CE, 3), (DE, 1), (DF, 9), (DG, 2), (EF, 4), (FH, 6), (GH, 5)\}$

MST $\rightarrow$ (using Boruvka's -)

• list final MST & total weight

AD
ED
BC
GH
FG

Explain why Boruvka's algorithm can be efficiently parallelized.
 (20) //

2) $V = \{A, B, C, D\}$ $E = \{(AB, 5), (AC, 9), (AD, \infty), (BC, 2), (BD, 6), (CD, 1), (DA, 3)\}$

$\rightarrow$ Directed. $\infty \rightarrow$ no edge.

i) Draw graph

ii) Find D^0 for Floyd Warshall

iii) Apply Floyd Warshall

3) let the A be the area of the region that lies under the graph of $f(x) = e^{-5x}$, between $x=0$, $x=2$.

a) Using the ^{right} endpoints, find an expression for A as a limit!

b) Evaluate it for 4 subintervals

c) Estimate the area by taking the sample points to be midpoints for 4 subintervals

d) Evaluate $\int_0^2 f(x) dx$ & justify which method is more approximate

3) The number of bacteria in a culture grows at a rate that is proportional to the number of present. Initially there were 10 bacteria in the culture. If doubling time of the culture is 3 hours,

a) Find the number of bacteria that were present after 24 hours. (2560 bacteria)

b) Determine the time period required for culture to reach 5000 bacteria.

$$3 \times \log_2 5000 \approx 26.90 \text{ hrs.}$$

1) $f(1) = 5$, $f(5) = 7$

$\lim_{x \rightarrow 3^+} f(x) = -\infty$ $\lim_{x \rightarrow 3^-} f(x) = -\infty$

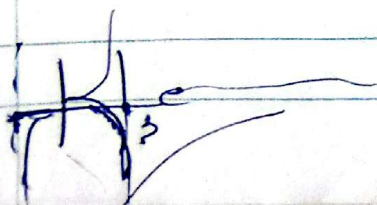
$$f'(3) = 0$$

$$f'(x) < 0 \text{ for } x > 5$$

$$f''(-1) = 0$$

$$f''(5) < 0$$

$$f''(x) > 0 \text{ for } x < -1$$



1) Prove that following limit does not exist

a) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt{2}x^2 - y}{x^2 - y^2}$

$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$